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**UNIVERSITY  
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AUSTRALIA**

## CSIT-26-S3-30: AI-Based Misinformation Detection for Short-Form Content

FYP-26-S3-30: User Requirement Specification

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# 1. Introduction

## 1.1 Purpose

This User Requirement Specification (URS) defines the user needs, operating context and system features for uiabo, an AI-assisted Android application that helps Singapore citizens aged 65 and above assess potentially misleading short-form online content. It translates the approved project scope into a common requirements baseline for design, implementation, testing and project acceptance.

The document identifies the intended user classes, describes the environment and constraints within which the application will operate, and records the user stories and use-case flows that the project team will implement.

uiabo is an AI-based misinformation tool used for decision-support. Its risk scores, likelihood scores, explanations and evidence links are intended to help users make a more informed judgement; they do not guarantee that content is true or false and do not replace professional fact-checkers.

## 1.2 Intended Audience and Reading Suggestions

The intended audience includes the five-person FYP development team, the project supervisor Mr Liaw Chun Huei and project assessor Mr Terrence Chew. The user-facing requirements are written in plain language so that readers do not require specialist knowledge of artificial intelligence or software engineering.

Readers seeking an overview should begin with Section 1 and Section 2. Readers reviewing behaviour for a particular user role should proceed to the relevant use-case diagram, user-story list and use-case descriptions in Section 3. The terms free user, premium user, system administrator and data engineer are used consistently throughout the document. Unless stated otherwise, “user” refers to an authenticated free or premium user.

## 1.3 Project Scope

The scope of uiabo is an elderly-friendly, English-language Android mobile application for Singapore citizens aged 65 and above. The initial prototype will support short-form text, supported public webpage or article links and static images. Users will receive evidence-backed assessments presented through clear concern labels, risk or likelihood scores, uncertainty information, plain-language explanations and links to supporting or contradicting sources.

The application will provide the following functions within the selected prototype scope:

- Free and premium account registration, authentication, password reset, profile management and account deletion.
- Text and public webpage-link submission, including malicious-link screening for supported links.
- Automated factual-claim identification, retrieval of previous fact-checks and relevant evidence, source assessment and evidence-linked result generation.

- A result page showing a concern label, misinformation-risk score, uncertainty, the identified claim, a plain-language explanation, citations and a recommended next action.
- Result history, remaining-allowance display, incorrect-result reporting, reviews, feedback and shareable result links.
- Premium image functions, including images with captions, OCR text review and correction, caption-image context analysis, and AI-generated or manipulated-image likelihood detection.
- Administrative account querying, account viewing, authorised operational-account creation, suspension and unsuspension, and feedback review.
- Data-engineering dashboards for reported incorrect results, report details and data-ingestion pipeline monitoring.

Access is tiered. A free user may complete one successful text or webpage-link submission per day. A premium user may complete up to 60 successful submissions per month and may access all supported image-analysis functions.

## 2. Overall Description

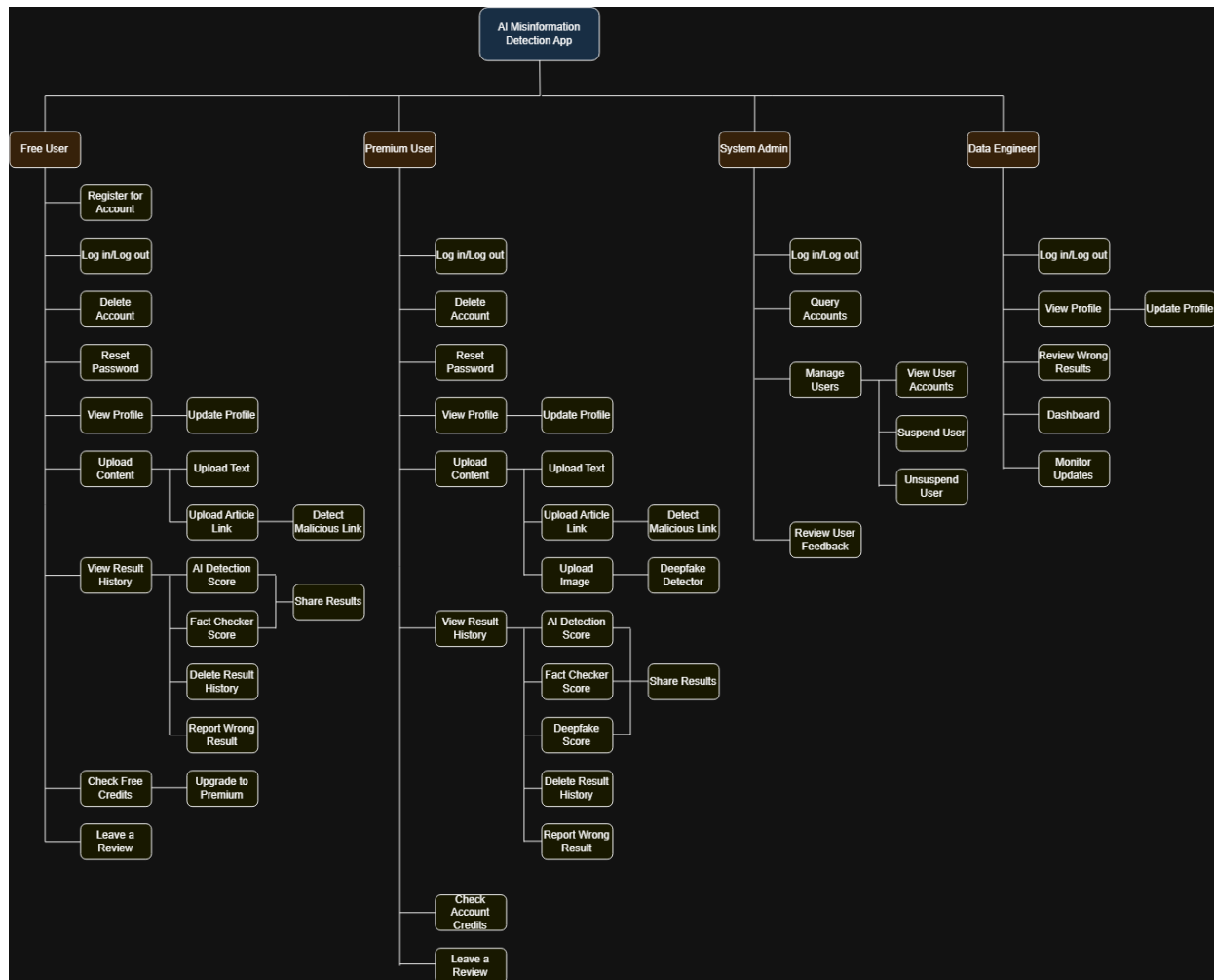
### 2.1 Product Perspective

uiabo is a Android mobile application that combines account management, content submission, AI-assisted analysis, evidence retrieval and explainable result presentation in one guided workflow. It is designed for older users who may be comfortable using smartphones and messaging applications but may have limited experience with formal fact-checking methods. The application does not replace professional fact-checkers; it helps users review evidence before believing or forwarding suspicious content.

The React Native mobile client will communicate over HTTPS with a Python FastAPI back end. Firebase Authentication will manage email-and-password authentication, while Cloud Firestore will store user profiles, roles, access tiers, allowances, submissions, results, evidence records, feedback and reported incorrect results. The back end will coordinate OCR, claim analysis, evidence retrieval, malicious-link screening and supported image-analysis services while keeping sensitive service credentials and analysis logic outside the mobile client.

Individual users are divided into free and premium tiers. Both tiers can check English text and supported public webpage links, review their results and evidence, manage their own accounts and provide feedback. Premium users additionally receive image, OCR, caption-image context and image-authenticity functions. Operational users are divided into system administrators, who manage account access and review feedback, and data engineers, who investigate reported incorrect results and monitor data-quality pipelines.

## 2.2 Product Features



## 2.2.1 Base Requirements

Base requirements are the minimum product features selected for the uiabo prototype. They cover account access, supported misinformation-analysis inputs, result access, tier management and the operational workflows required to administer and improve the service.

| Feature ID | User class           | Product feature                                                                                                                                 | Related use case(s)                 |
|------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PF-B01     | All four roles       | Log in and log out; reset password; view and update the authenticated user's own profile.                                                       | UC-01–02, 04–06; UC-25–29; UC-36–40 |
| PF-B02     | Free and Premium     | Register for a free or premium account and permanently delete the individual user's own account.                                                | UC-03, UC-16, UC-18                 |
| PF-B03     | Free and Premium     | Submit English short-form text for claim identification, evidence retrieval and misinformation-risk analysis.                                   | UC-07                               |
| PF-B04     | Free and Premium     | Submit a supported public webpage or article link for misinformation analysis and malicious-link screening.                                     | UC-08                               |
| PF-B05     | Free and Premium     | View the personal results page and individual result details, view the remaining allowance and report a result that appears incorrect.          | UC-09–11, UC-13                     |
| PF-B06     | Free User            | Upgrade a free account to premium access.                                                                                                       | UC-17                               |
| PF-B07     | Premium User         | Renew or cancel the monthly premium subscription.                                                                                               | UC-19–20                            |
| PF-B08     | Premium User         | Upload a supported image with a caption for combined image-and-caption analysis.                                                                | UC-21                               |
| PF-B09     | Premium User         | Extract visible text from an image using OCR, allow the user to review and correct it, and analyse the corrected text.                          | UC-23                               |
| PF-B10     | Premium User         | Analyse whether an image and its caption are contextually consistent with respect to cues such as date, location, event or description.         | UC-22                               |
| PF-B11     | Premium User         | Assess whether a supported image may be AI-generated or manipulated and present the likelihood, uncertainty and available forensic explanation. | UC-24                               |
| PF-B12     | System Administrator | Query for an account and view the account details permitted for administration.                                                                 | UC-30–31                            |
| PF-B13     | System Administrator | Create an authorised operational account for a new system administrator or data engineer.                                                       | UC-32                               |
| PF-B14     | System Administrator | Suspend or unsuspend an eligible account without deleting the account record.                                                                   | UC-33–34                            |
| PF-B15     | Data Engineer        | View the incorrect-results dashboard and inspect the details of reported incorrect results.                                                     | UC-41–42                            |
| PF-B16     | Data Engineer        | Monitor data-ingestion pipelines to identify failed, delayed or incomplete updates.                                                             | UC-43                               |

## 2.2.2 Additional Requirements

Additional requirements are useful quality-of-life and feedback features that support users and the operational team but are not required for the minimum misinformation-analysis workflow.

| Feature ID | User class           | Product feature                                                                                 | Related use case(s) |
|------------|----------------------|-------------------------------------------------------------------------------------------------|---------------------|
| PF-A01     | Free and Premium     | Delete selected result-history records to remove unwanted items from the personal results page. | UC-12               |
| PF-A02     | Free and Premium     | Leave a review describing the user's experience with uiabo.                                     | UC-14               |
| PF-A03     | Free and Premium     | Generate and share a result through a link.                                                     | UC-15               |
| PF-A04     | System Administrator | View user reviews and other feedback from the administrative feedback page.                     | UC-35               |

## 2.2.3 Functional Feature Behaviour

The following summary explains what the principal feature groups do. Detailed actor interactions, pre-conditions, flows and failure handling are specified in Section 3.

| Feature area                           | Functional behaviour                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Account access and management</b>   | Firebase Authentication will support email-and-password registration, login, logout and password reset. Authenticated users can view and update their own profiles. Individual users can permanently delete their own accounts, while protected roles and access tiers are enforced by the back end.                                                     |
| <b>Text analysis</b>                   | A free or premium user can paste English short-form text. The system identifies objectively checkable claims, distinguishes non-verifiable content where possible, retrieves relevant evidence and generates an evidence-linked result.                                                                                                                  |
| <b>Link analysis</b>                   | A free or premium user can submit a supported public webpage or article link. The system screens the URL for malicious-link indicators, extracts accessible content, analyses checkable claims and presents the available evidence and source information.                                                                                               |
| <b>Image and caption analysis</b>      | A premium user can upload a supported static image with an English caption. The system analyses the combined content and presents a risk assessment, uncertainty, explanation, evidence and recommended next action.                                                                                                                                     |
| <b>OCR text analysis</b>               | A premium user can extract visible text from an image, review and correct the OCR output, and submit the corrected text for misinformation analysis.                                                                                                                                                                                                     |
| <b>Caption-image context analysis</b>  | A premium user can ask the system to compare an image with its caption and identify possible contextual mismatches involving the date, location, event or description. Insufficient evidence produces an inconclusive result rather than a definitive accusation.                                                                                        |
| <b>Deepfake/AI image analysis</b>      | A premium user can request a probabilistic AI-generation or manipulation likelihood assessment. The result explains the available detector signals and limitations in plain language and does not present an individual artefact or score as proof.                                                                                                      |
| <b>Results, allowance and feedback</b> | Successful analyses are saved to the user's personal results page and consume the applicable allowance: one successful text or link submission per day for free users, or up to 60 successful submissions per month for premium users. Users can inspect evidence, report an incorrect result and use the selected history, review and sharing features. |

| Feature area                 | Functional behaviour                                                                                                                                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Operational functions</b> | System administrators manage authorised account access and feedback. Data engineers review reported incorrect results, related provenance and pipeline status so model, evidence or data-quality problems can be investigated. |

## 2.3 User Classes and Characteristics

| User class           | Characteristics and permitted functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Free User            | A registered individual user, primarily a Singapore citizen aged 65 or above, who can use an Android smartphone and benefits from a simple, guided interface. The free user can manage their account and profile; submit English short-form text or a supported public webpage link; receive evidence-backed results; view one daily successful-submission allowance; manage result history; report an incorrect result; leave a review; share a result; and upgrade to premium. Image, OCR, caption-image context and deepfake/AI-image functions are unavailable on the free tier. |
| Premium User         | A registered individual user with an active monthly premium subscription. The premium user retains all free-user functions, receives up to 60 successful submissions per month and can upload static images with captions, review and correct OCR-extracted text, analyse caption-image context and request an AI-generated or manipulated-image likelihood assessment. The user can renew or cancel the subscription and is expected to understand that all scores are probabilistic decision support.                                                                              |
| System Administrator | An authorised operational user who manages access to the application. The system administrator can authenticate, manage their own profile, query accounts, view the permitted account details, create authorised operational accounts, suspend or unsuspend accounts and review user feedback. Administrative access is role-restricted, logged and limited to the information required for the task.                                                                                                                                                                                |
| Data Engineer        | An authorised operational user responsible for monitoring the information and model-support workflow. The data engineer can authenticate, manage their profile, view a dashboard of reported incorrect results, inspect individual reports and monitor data-ingestion pipelines to identify failed or incomplete updates. The role requires technical knowledge of data quality, provenance and model evaluation.                                                                                                                                                                    |

## 2.4 Operating Environment

| Environment      | Requirement                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobile client    | A supported Android smartphone capable of running the React Native with Expo build. The initial prototype is intended for Android and is not required to provide an iOS or desktop client.                                                     |
| Connectivity     | A stable internet connection is required for authentication, uploads, cloud data access, evidence retrieval and AI-assisted processing. The application should handle temporary loss of connectivity without exposing or corrupting user data. |
| Back-end runtime | A Python environment running FastAPI through Uvicorn, exposed through authenticated HTTPS REST endpoints and capable of coordinating the selected analysis services.                                                                           |
| Cloud services   | Firebase Authentication and Cloud Firestore must be reachable and correctly configured for authentication, role checks, account data, usage allowances, submissions, results and feedback records.                                             |

| Environment          | Requirement                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content and language | The initial prototype processes English-language short-form text, supported public webpage/article links and supported static-image formats. Video, audio and non-English analysis are outside the selected scope. |
| User interface       | The mobile interface should support readable text, clear contrast, large touch targets, simple navigation and plain-language messages suitable for the target group of users aged 65 and above.                    |

## 2.5 Design and Implementation Constraints

| Constraint              | Description                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Platform                | The prototype will be delivered as an Android application using JavaScript, React Native and Expo. Functions that depend on unsupported platforms or device capabilities are excluded.                                                                                 |
| Back-end                | Server-side processing and AI/data integration will use Python, FastAPI and Uvicorn. The mobile application will access the back end through HTTPS REST endpoints using JSON and Axios.                                                                                |
| Data and authentication | Cloud Firestore and Firebase Authentication are the selected database and authentication services. Access rules and back-end role checks must prevent a user from accessing another user's protected information or operational functions.                             |
| Access tiers            | Free access is limited to one successful text or webpage-link submission per day. Premium access is limited to 60 successful submissions per month and unlocks the supported image functions. Failed, cancelled or timed-out analyses should not consume an allowance. |
| Supported inputs        | The prototype is limited to English short-form text, supported public webpage/article links and static images. Private pages, paywalled or inaccessible content, video and speech processing are outside scope.                                                        |
| Explainability          | Results must not be presented as mathematical proof. The interface must communicate uncertainty, provide a Not Enough Information outcome when appropriate and distinguish misinformation risk from the separate image-generation or manipulation likelihood score.    |
| Accessibility           | The interface must prioritise elderly-friendly interaction through readable typography, large touch targets, clear instructions, limited cognitive load and understandable error and result messages.                                                                  |
| Security                | Sensitive user data must be protected in transit and at rest. Authentication is required for protected functions, sessions expire after 30 minutes of inactivity, and third-party service credentials must remain on the server.                                       |

## 2.6 User Documentation

The application will provide concise onboarding and in-app guidance written in plain English. Guidance will explain how to register or log in, choose a submission type, paste text or a public

link, upload a supported image, add or correct captions and OCR text, submit content, read a result, inspect evidence links and decide whether content should be forwarded. Premium-only functions and the user's remaining allowance will be clearly identified before submission.

Result guidance will explain the meanings of Low Concern, Needs Caution, High Concern and Not Enough Information; the difference between misinformation risk and image-generation or manipulation likelihood; and the limitations of probabilistic AI output. The interface will also provide instructions for reporting an incorrect result, leaving a review, sharing a result link, managing history, resetting a password and changing subscription status. Error messages will state what happened and the next safe action, such as correcting an input, retrying later or checking the network connection.

## 2.7 Assumptions and Dependencies

| Assumption or dependency            | Description                                                                                                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assumption 1 – User access          | Individual users have access to a supported Android smartphone, a registered email address and a sufficiently stable internet connection. A family member or community helper may assist with onboarding when needed.                         |
| Assumption 2 – Submitted content    | Users submit content they are permitted to process and provide sufficiently clear English text, accessible public webpage links or supported static images. Users do not rely on uiabo as the sole basis for a high-impact decision.          |
| Assumption 3 – Account information  | Users provide accurate registration and profile information, protect their credentials and review confirmation prompts before permanent account or history deletion.                                                                          |
| Assumption 4 – Evidence limitations | Relevant and credible online evidence may not exist for every claim. When evidence is weak, conflicting or unavailable, the system may return Not Enough Information rather than a confident assessment.                                      |
| Dependency 1 – Cloud platform       | Authentication, profile, allowance, submission, result and feedback functions depend on the continued availability and correct configuration of Firebase Authentication and Cloud Firestore.                                                  |
| Dependency 2 – Analysis services    | OCR, AI models, malicious-link screening, fact-check retrieval, web search and image-authenticity functions depend on selected internal or third-party services, their rate limits, supported formats and available output.                   |
| Dependency 3 – External sources     | Evidence quality and result generation depend on access to current authoritative, academic, news and fact-check sources. Source pages may change, become unavailable or restrict automated access.                                            |
| Dependency 4 – Subscription service | Premium activation, renewal and cancellation depend on the selected payment or app-store subscription mechanism and successful confirmation of the account's entitlement.                                                                     |
| Dependency 5 – Performance          | End-to-end response time depends on the user's network, upload size, cloud services and analysis providers. The system must time out processing that exceeds 20 minutes and clearly preserve the user's allowance when a run is unsuccessful. |

### **3. System Features**

This section describes the user-visible and operational behaviour of uiabo. Use-case identifiers UC-01 to UC-43 follow the agreed user-story list. Shared individual-user use cases retain a single identifier and apply to both free and premium users; premium-specific use cases are listed separately.

### 3.1 Use Case Diagram – Free User

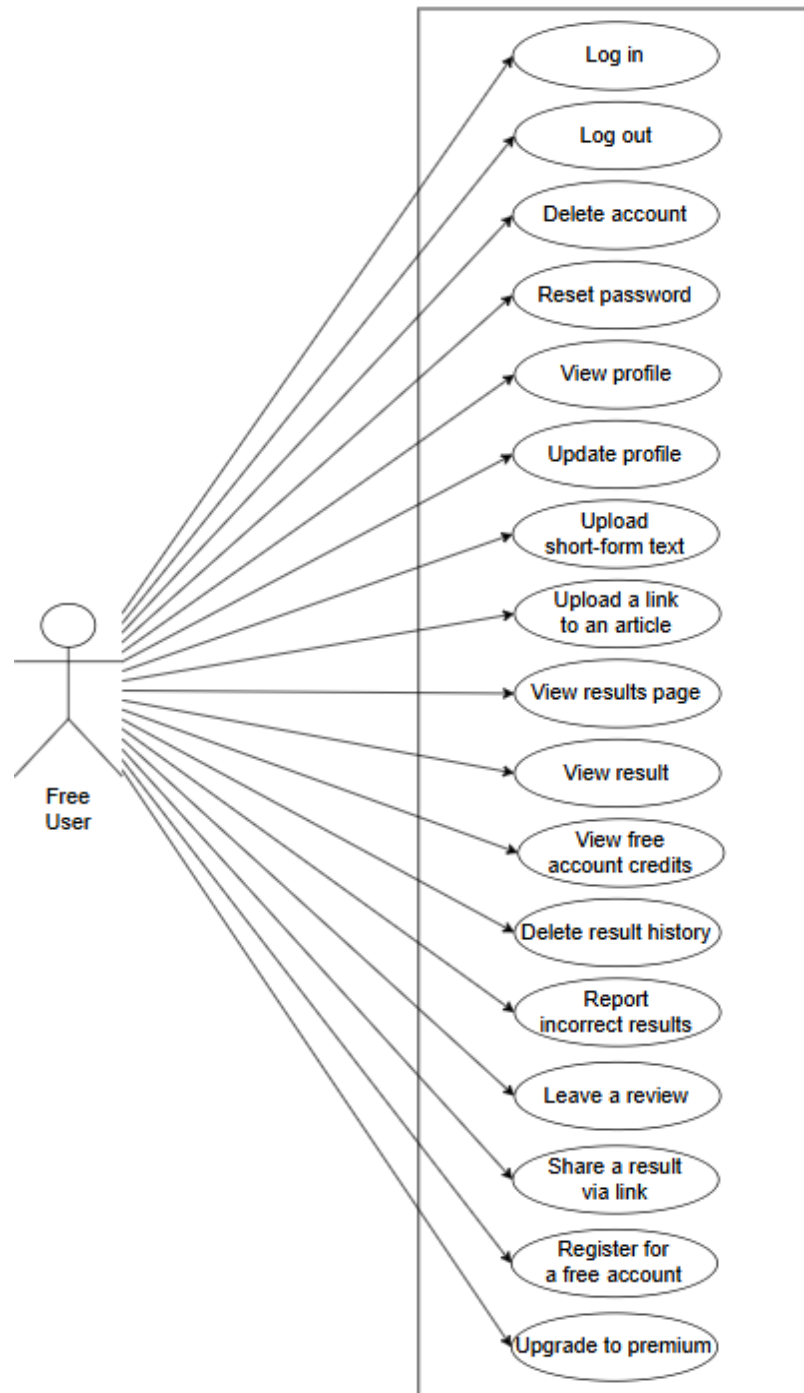


Figure 3.1. Free User Use Case Diagram

### 3.1.1 User Stories – Free User

- UC-01: As a free user, I want to log in to my account so that I can access the functions of the app.
- UC-02: As a free user, I want to log out of my account so that I can maintain the security of my account.
- UC-03: As a free user, I want to delete my account so that I can remove my personal data from the system.
- UC-04: As a free user, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-05: As a free user, I want to view my profile so that I can check whether my account information is up to date.
- UC-06: As a free user, I want to update my profile so that I can keep my account information up to date.
- UC-07: As a free user, I want to upload short-form text so that I can find out whether the submitted text contains misinformation.
- UC-08: As a free user, I want to upload a link to an article so that I can find out whether a linked article or webpage contains misinformation.
- UC-09: As a free user, I want to view my personal results page so that I can access the results of my submitted content.
- UC-10: As a free user, I want to view the result of my submitted content so that I can view the details of a specific analysis result.
- UC-11: As a free user, I want to view my remaining account credits so that I can find out how many uses of the app I have left.
- UC-12: As a free user, I want to delete my result history so that I can remove results I no longer want to see from my results page.
- UC-13: As a free user, I want to report results that are incorrect so that I can provide data and feedback so the system can be improved in future.
- UC-14: As a free user, I want to leave a review so that I can express my experience using the app so it can be improved in the future.
- UC-15: As a free user, I want to share a result via link so that I can inform others of content containing misinformation.
- UC-16: As a free user, I want to register for a free account so that I can gain access to the free functions of the app.
- UC-17: As a free user, I want to upgrade to premium so that I can access all the functions of the app and gain more uses of the functions.

### 3.1.2 Use Case Description – Free User

Use Cases 01–15 apply to both free and premium users. Use Cases 16–17 apply only to free users.

## Use Case 01 – User logs in to their account

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User logs in to their account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ID:</b> 01 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Access the functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Description:</b> A registered free or premium user authenticates with their email address and password to access the personalised functions and data for their account.                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The user selects the Log In button on the login screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Pre-condition:</b> The user has a registered, active account, is not already logged in, and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Main Flow:</b> 1. The user opens uiabo and selects Log In.<br>2. The system displays the login form.<br>3. The user enters their registered email address and password.<br>4. The user selects Log In.<br>5. The system validates the submitted credentials using Firebase Authentication.<br>6. The system retrieves the user's role, access tier and account status.<br>7. The system creates the authenticated session and displays the user's personalised home screen.                                                                                     |               |
| <b>Sub-flow:</b> S1. If the user has forgotten their password, they select Forgot Password and continue with Use Case 04.                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Alternative flow:</b> 5a. If the email address or password is incorrect, the system does not create a session and displays a clear error message.<br>5b. If required information is missing, the system highlights the incomplete field and asks the user to correct it.<br>6a. If the account is suspended, the system denies access and informs the user that the account is unavailable.<br>7a. If authentication or account data cannot be retrieved because of a network or service error, the system keeps the user logged out and offers a retry option. |               |

## Use Case 02 – User logs out of their account

|                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User logs out of their account                                                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 02 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Maintain the security of their account.                                                                                                                                                                                                                                                                                                                 |               |
| <b>Description:</b> An authenticated user ends the current uiabo session so that account functions and personal data are no longer accessible from the device without logging in again.                                                                                                                                                                                                                            |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The user selects Log Out from the account or profile menu.                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Pre-condition:</b> The user is logged in to an active uiabo session.                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Main Flow:</b> 1. The user opens the account or profile menu.<br>2. The user selects Log Out.<br>3. The system asks the user to confirm the action.<br>4. The user confirms that they want to log out.<br>5. The system ends the authenticated session and removes locally stored session credentials.<br>6. The system returns the user to the login screen and protected account data is no longer displayed. |               |
| <b>Sub-flow:</b> S1. After 30 minutes of inactivity, the system automatically expires the session, clears access to protected screens and returns the user to the login screen.                                                                                                                                                                                                                                    |               |
| <b>Alternative flow:</b> 4a. If the user cancels the confirmation, the system closes the prompt and keeps the session active.<br>5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and informs the user that the server could not be reached.                                                                                                             |               |

### Use Case 03 – User deletes their account

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User deletes their account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ID:</b> 03 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Remove their personal data from the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Description:</b> A free or premium user permanently deletes their uiabo account, associated profile information and saved result history. The action is irreversible.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Trigger:</b> The user selects Delete Account from the account or profile settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is logged in, can re-authenticate when requested, and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The user opens the account or profile settings.<br>2. The user selects Delete Account.<br>3. The system explains that account data and saved result history will be permanently deleted and cannot be recovered.<br>4. The system asks the user to confirm the deletion and re-authenticate.<br>5. The user provides the required authentication and confirms Delete Permanently.<br>6. The system validates the user's identity.<br>7. The system deletes the user's account record, personal data and associated result history, then removes the authentication account.<br>8. The system ends the session, confirms deletion and returns to the welcome or login screen. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Alternative flow:</b> 5a. If the user cancels, the system closes the confirmation and does not delete any data.<br>6a. If re-authentication fails, the system refuses the deletion and asks the user to try again or reset their password.<br>7a. If any required deletion operation fails, the system does not report the account as deleted, records the failure safely and offers a retry or support option.<br>8a. If the network connection is lost, the system informs the user that deletion could not be completed and keeps the account accessible until deletion succeeds.                                                                                                           |               |

## Use Case 04 – User resets their password

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| <b>Name:</b> User resets their password                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ID:</b> 04 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Improve account security or regain access when the password is forgotten.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Description:</b> A free or premium user requests a secure password-reset link and replaces the existing password with a new valid password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Trigger:</b> The user selects Forgot Password on the login screen or Reset Password in account settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Pre-condition:</b> The user has a registered email-and-password account, can access the registered email inbox, and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Main Flow:</b> 1. The user selects Forgot Password or Reset Password.<br>2. The system displays the password-reset form.<br>3. The user enters their registered email address and submits the request.<br>4. The system validates the request and sends a time-limited reset link to the registered email address.<br>5. The system displays a neutral confirmation instructing the user to check their email.<br>6. The user opens the reset link and enters the new password twice.<br>7. The system checks that the reset link is valid and that the new password meets the password rules.<br>8. The system updates the password through Firebase Authentication and invalidates the reset link.<br>9. The system confirms the reset and directs the user to log in with the new password. |               |
| <b>Sub-flow:</b> S1. A logged-in user may start the same reset process from account settings as a preventative security measure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Alternative flow:</b> 3a. If the email field is empty or invalid, the system highlights the field and does not submit the request.<br>4a. To avoid revealing whether an account exists, the system displays the same neutral confirmation even when no matching account can be disclosed.<br>7a. If the reset link is invalid, expired or already used, the system rejects it and offers to send a new link.<br>7b. If the passwords do not match or the new password fails the password rules, the system explains the problem and asks the user to correct it.<br>8a. If the service or network is unavailable, the system does not change the password and offers a retry option.                                                                                                           |               |

## Use Case 05 – User views their profile

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| <b>Name:</b> User views their profile                                                                                                                                                                                                                                                                                                                                                                               | <b>ID:</b> 05 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Check whether their account information is up to date.                                                                                                                                                                                                                                                                                                   |               |
| <b>Description:</b> A logged-in free or premium user views the profile information stored for their own account.                                                                                                                                                                                                                                                                                                    |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Trigger:</b> The user selects Profile from the navigation or account menu.                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Pre-condition:</b> The user is logged in to an active account.                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Main Flow:</b> 1. The user opens the navigation or account menu.<br>2. The user selects Profile.<br>3. The system verifies the current session and retrieves only the authenticated user's profile information.<br>4. The system displays the profile information and the user's account tier in a readable format.<br>5. The user reviews the displayed information.                                            |               |
| <b>Sub-flow:</b> S1. The user may select Edit Profile and continue with Use Case 06.                                                                                                                                                                                                                                                                                                                                |               |
| <b>Alternative flow:</b> 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again.<br>3b. If the profile cannot be retrieved, the system displays an error without exposing another user's data and offers a retry option.<br>4a. If an optional profile field has no value, the system shows it as not provided rather than displaying incorrect information. |               |

## Use Case 06 – User updates their profile

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| <b>Name:</b> User updates their profile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>ID:</b> 06 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Keep their account information up to date.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Description:</b> A logged-in free or premium user changes editable personal information in their own profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The user selects Edit Profile while viewing their profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Pre-condition:</b> The user is logged in, has opened their profile, and the existing profile information is available.                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Main Flow:</b> 1. The user views their profile and selects Edit Profile.<br>2. The system displays the editable profile fields with their current values.<br>3. The user changes one or more fields.<br>4. The user selects Save Changes.<br>5. The system validates the updated information.<br>6. The system saves the valid changes to the authenticated user's profile.<br>7. The system displays the updated profile and confirms that the changes were saved.                                                                                             |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Alternative flow:</b> 4a. If the user selects Cancel, the system discards unsaved changes and returns to the profile view.<br>5a. If required information is missing or has an invalid format, the system highlights the relevant field and does not save the changes.<br>5b. If a changed unique value is already used by another account, the system asks the user to provide a different value.<br>6a. If the update fails because of a network, authentication or database error, the system retains the previous stored profile and offers a retry option. |               |

## Use Case 07 – User uploads short-form text

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| <b>Name:</b> User uploads short-form text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ID:</b> 07 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Find out whether the submitted text contains misinformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Description:</b> A free or premium user pastes English-language short-form text for automated claim identification, evidence retrieval and misinformation-risk analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The user selects the short-form text analysis option and submits text.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is logged in, has an internet connection and has at least one successful submission remaining in the applicable allowance (one per day for free users or up to 60 per month for premium users).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Main Flow:</b> 1. The user selects the short-form text analysis option.<br>2. The system displays a text-entry screen and the user's remaining submission allowance.<br>3. The user pastes or enters the short-form text.<br>4. The user selects Analyse.<br>5. The system validates that the text is present and supported and confirms that an allowance remains.<br>6. The system creates the analysis request and shows clear progress information.<br>7. The system identifies the principal objectively checkable factual claim or claims.<br>8. The system searches recognised fact-checking and credible sources for supporting or contradicting evidence.<br>9. The system generates the concern label, misinformation-risk score, uncertainty information, plain-language explanation and evidence links.<br>10. The system saves the completed result to the user's result history and deducts one successful submission from the applicable allowance.<br>11. The system displays the completed result as described in Use Case 10. |               |
| <b>Sub-flow:</b> S1. If no objectively checkable claim is found, the system creates a Not Enough Information result with a plain-language explanation rather than labelling opinion, humour or an unverifiable statement as misinformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Alternative flow:</b> 5a. If the text is empty, the system asks the user to enter text and does not start or deduct an analysis.<br>5b. If the text is unsupported or cannot be processed, the system explains the issue and allows the user to edit the submission.<br>5c. If no allowance remains, the system does not start the analysis, displays when access becomes available again and may offer the free user an upgrade option.<br>8a. If evidence services are temporarily unavailable, the system reports that the analysis could not be completed and offers a retry option.<br>9a. If processing exceeds the 20-minute timeout, the system cancels the request, informs the user and does not deduct a successful submission.                                                                                                                                                                                                                                                                                                      |               |

## Use Case 08 – User uploads a link to an article

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| <b>Name:</b> User uploads a link to an article                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ID:</b> 08 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Find out whether a linked article or webpage contains misinformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Description:</b> A free or premium user submits a supported public webpage or article URL. uiabo screens the URL for malicious-link indicators, extracts available article content and analyses checkable claims.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Trigger:</b> The user selects the link analysis option and submits a webpage or article URL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Pre-condition:</b> The user is logged in, has an internet connection and has at least one successful submission remaining in the applicable daily or monthly allowance. The target URL is intended to be public.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <p><b>Main Flow:</b></p> <ol style="list-style-type: none"> <li>1. The user selects the link analysis option.</li> <li>2. The system displays a URL-entry screen and the user's remaining submission allowance.</li> <li>3. The user enters or pastes a public webpage or article URL.</li> <li>4. The user selects Analyse.</li> <li>5. The system validates the URL format, supported protocol and remaining allowance.</li> <li>6. The system screens the submitted URL for malicious-link indicators before retrieving article content.</li> <li>7. The system safely retrieves and extracts the available text and relevant source information from the public page.</li> <li>8. The system identifies the principal objectively checkable factual claim or claims.</li> <li>9. The system searches recognised fact-checking and credible sources for supporting or contradicting evidence.</li> <li>10. The system generates the concern label, misinformation-risk score, uncertainty information, plain-language explanation, evidence links and available source context.</li> <li>11. The system saves the completed result and deducts one successful submission from the applicable allowance.</li> <li>12. The system displays the completed result as described in Use Case 10.</li> </ol> |               |
| <p><b>Sub-flow:</b></p> <p>S1. If malicious-link indicators are detected, the system displays a prominent warning and does not open the submitted link automatically. Analysis continues only when content can be retrieved safely.</p> <p>S2. If no objectively checkable claim is found, the system creates a Not Enough Information result with an explanation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <p><b>Alternative flow:</b></p> <p>5a. If the URL is empty, malformed or uses an unsupported protocol, the system explains the required format and does not start or deduct an analysis.</p> <p>5b. If no allowance remains, the system does not start the analysis, displays when access becomes available again and may offer the free user an upgrade option.</p> <p>7a. If the page is private, inaccessible, unsupported, blocked or contains no extractable content, the system explains that it cannot analyse the page and does not deduct a successful submission.</p> <p>9a. If evidence services are temporarily unavailable, the system reports that the analysis could not be completed and offers a retry option.</p> <p>10a. If processing exceeds the 20-minute timeout, the system cancels the request, informs the user and does not deduct a successful submission.</p>                                                                                                                                                                                                                                                                                                                                                                                                               |               |

## Use Case 09 – User views their personal results page

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| <b>Name:</b> User views their personal results page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>ID:</b> 09 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Access the results of their submitted content.                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Description:</b> A logged-in free or premium user views the saved analysis results associated with their own account.                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Trigger:</b> The user selects Results or Result History from the app navigation.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Pre-condition:</b> The user is logged in to an active account.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Main Flow:</b> 1. The user opens the app navigation.<br>2. The user selects Results or Result History.<br>3. The system verifies the current session.<br>4. The system retrieves only the result records associated with the authenticated account.<br>5. The system displays the user's saved results in a readable list, including a useful summary such as submission type, date, concern label and analysis status.<br>6. The user reviews the list of results.<br>7. The user may select a completed result to continue with Use Case 10. |               |
| <b>Sub-flow:</b> S1. If the account has no saved results, the system displays an empty-state message and a clear option to submit content for analysis.<br>S2. If an analysis is still processing, the system displays its current status without presenting an unfinished assessment as final.                                                                                                                                                                                                                                                   |               |
| <b>Alternative flow:</b> 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again.<br>4a. If the results cannot be retrieved, the system displays an error, does not expose another user's results and offers a retry option.<br>5a. If an individual result was deleted, it is no longer displayed in the list.                                                                                                                                                                             |               |

## Use Case 10 – User views the result of their submitted content

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| <b>Name:</b> User views the result of their submitted content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ID:</b> 10 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – View the details of a specific analysis result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Description:</b> A free or premium user opens a completed result to review the system's assessment, reasoning, uncertainty and evidence for the submitted content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Trigger:</b> The user selects a completed result from the personal results page, or the system completes a newly submitted analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is logged in, the result belongs to the authenticated user, and the analysis has completed successfully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <p><b>Main Flow:</b></p> <ol style="list-style-type: none"> <li>1. The user selects a completed result, or a new analysis finishes.</li> <li>2. The system verifies that the authenticated user is authorised to view the result.</li> <li>3. The system retrieves the saved submission, assessment and evidence records.</li> <li>4. The system displays the concern label: Low Concern, Needs Caution, High Concern or Not Enough Information.</li> <li>5. The system displays the misinformation-risk score, uncertainty information, identified factual claim and a plain-language explanation.</li> <li>6. The system displays supporting and contradicting evidence with source links or citations.</li> <li>7. Where applicable to a premium image result, the system separately displays the AI-generation or manipulation likelihood score and its limitations.</li> <li>8. The system displays an appropriate next-action recommendation and keeps the result available in the user's history.</li> </ol> |               |
| <p><b>Sub-flow:</b></p> <p>S1. The user may select an evidence link to inspect the source independently; the external source opens without replacing the saved result.</p> <p>S2. If the result is Not Enough Information, the system explains that insufficient checkable evidence was found and does not present a definitive misinformation judgement.</p> <p>S3. The user may use the separate report, share, review or delete-result functions when those actions are available.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <p><b>Alternative flow:</b></p> <p>2a. If the result belongs to another account, the system denies access and does not reveal the result data.</p> <p>3a. If the result was deleted or cannot be found, the system informs the user and returns to the personal results page.</p> <p>3b. If the result cannot be loaded because of a network or database error, the system offers a retry option.</p> <p>6a. If an external evidence page is no longer available, the system keeps the saved assessment visible and marks the affected link as unavailable when detected.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |

## Use Case 11 – User views their free account credits

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| <b>Name:</b> User views their free account credits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ID:</b> 11 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Find out how many uses of the app they have left.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Description:</b> A logged-in free or premium user views the number of analysis submissions remaining in their applicable allowance and when that allowance next refreshes.                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Trigger:</b> The user selects Credits from the account or profile menu, or views the allowance indicator shown on a submission screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Pre-condition:</b> The user is logged in to an active account.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Main Flow:</b> 1. The user opens the account or profile menu.<br>2. The user selects Credits.<br>3. The system verifies the current session.<br>4. The system retrieves the authenticated user's account tier and submission usage.<br>5. The system displays the remaining allowance, the applicable limit (one successful submission per day for free users or up to 60 successful submissions per month for premium users) and the date on which the allowance next refreshes.<br>6. The user reviews the displayed allowance.                                                         |               |
| <b>Sub-flow:</b> S1. The system also displays the remaining allowance on the text and link submission screens.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Alternative flow:</b> 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again.<br>4a. If the usage record cannot be retrieved, the system displays an error rather than an inaccurate balance and offers a retry option.<br>5a. If a submission is still processing, the system shows it as pending and does not deduct it from the allowance until the analysis completes successfully.<br>5b. If an analysis was cancelled or timed out, the system does not deduct it from the allowance and the displayed balance is unchanged. |               |

## Use Case 12 – User deletes their result history

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| <b>Name:</b> User deletes their result history                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ID:</b> 12 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Remove results they no longer want to see from their results page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Description:</b> A logged-in free or premium user permanently deletes one or more saved analysis results from their own result history. The action is irreversible.                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Trigger:</b> The user selects Delete on a saved result, or selects Delete Selected on the personal results page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is logged in and has at least one saved result associated with the authenticated account.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Main Flow:</b> 1. The user opens their personal results page.<br>2. The user selects one or more saved results.<br>3. The user selects Delete.<br>4. The system explains that the selected results will be permanently deleted and cannot be recovered.<br>5. The user confirms the deletion.<br>6. The system verifies that the selected results belong to the authenticated account.<br>7. The system deletes the selected result records, including the stored submission, assessment and evidence records.<br>8. The system confirms the deletion and displays the updated results list. |               |
| <b>Sub-flow:</b> S1. The user may delete a single result while viewing it, after which the system returns to the personal results page.<br>S2. If the account has no saved results, the system displays an empty-state message and does not offer a delete option.                                                                                                                                                                                                                                                                                                                              |               |
| <b>Alternative flow:</b> 5a. If the user cancels, the system closes the confirmation and does not delete any data.<br>6a. If a selected result does not belong to the authenticated account, the system denies the deletion and does not reveal the result data.<br>7a. If the deletion fails because of a network or database error, the system does not report the results as deleted and offers a retry option.<br>7b. Deleting a result does not restore a used submission to the user's allowance.                                                                                         |               |

### Use Case 13 – User reports results that are incorrect

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| <b>Name:</b> User reports results that are incorrect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>ID:</b> 13 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Provide data and feedback so the system can be improved in future.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Description:</b> A free or premium user flags a completed result they believe is wrong and submits a reason so that the result can be reviewed by the data engineer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Trigger:</b> The user selects Report Incorrect Result while viewing a completed result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Pre-condition:</b> The user is logged in, is viewing a completed result that belongs to the authenticated account, and has not already reported that result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Main Flow:</b> 1. The user opens a completed result.<br>2. The user selects Report Incorrect Result.<br>3. The system displays the report form with a list of reasons and an optional comment field.<br>4. The user selects a reason and may enter additional comments.<br>5. The user submits the report.<br>6. The system validates the submitted report.<br>7. The system saves the report together with a reference to the result, the identified factual claim, the concern label and the misinformation-risk score that were produced.<br>8. The system confirms that the report has been submitted for review and marks the result as reported. |               |
| <b>Sub-flow:</b> S1. The submitted report is made available to the data engineer for review.<br>S2. Reporting a result does not change the displayed assessment, and the result remains visible in the user's history.                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Alternative flow:</b> 4a. If no reason is selected, the system highlights the field and does not submit the report.<br>6a. If the user has already reported the result, the system informs the user and does not create a duplicate report.<br>7a. If the report cannot be saved because of a network or database error, the system does not confirm submission and offers a retry option.<br>8a. If the user deletes the result after reporting it, the system retains only the information needed for review and does not retain the deleted personal result record.                                                                                 |               |

**Use Case 14 – User leaves a review**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User leaves a review                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ID:</b> 14 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Express their experience using the app so it can be improved in the future.                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Description:</b> A logged-in free or premium user submits a rating and written feedback about their overall experience with uiabo.                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Trigger:</b> The user selects Leave a Review from the account menu or the feedback page.                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Pre-condition:</b> The user is logged in to an active account.                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Main Flow:</b> 1. The user opens the account menu or the feedback page.<br>2. The user selects Leave a Review.<br>3. The system displays the review form with a rating scale and a comment field.<br>4. The user selects a rating and may enter comments.<br>5. The user submits the review.<br>6. The system validates the submitted review.<br>7. The system saves the review against the authenticated account.<br>8. The system confirms that the review has been submitted. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Alternative flow:</b> 4a. If no rating is selected, the system highlights the rating field and does not submit the review.<br>6a. If the comment exceeds the supported length, the system explains the limit and asks the user to shorten it.<br>7a. If the review cannot be saved because of a network or database error, the system does not confirm submission and offers a retry option.                                                                                     |               |

## Use Case 15 – User shares a result via link

|                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User shares a result via link                                                                                                                                                                                                                                                                                                                                                                      | <b>ID:</b> 15 |
| <b>Stakeholders &amp; Goals:</b> Free user, Premium user – Inform others of content containing misinformation.                                                                                                                                                                                                                                                                                                  |               |
| <b>Description:</b> A free or premium user generates a shareable link to a completed result so that other people can view the assessment without logging in to uiabo.                                                                                                                                                                                                                                           |               |
| <b>Actor(s):</b> Free user, Premium user                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Trigger:</b> The user selects Share while viewing a completed result.                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Pre-condition:</b> The user is logged in, and the result is completed and belongs to the authenticated account.                                                                                                                                                                                                                                                                                              |               |
| <b>Main Flow:</b> 1. The user opens a completed result.<br>2. The user selects Share.<br>3. The system explains what the shared page will display and asks the user to confirm.<br>4. The user confirms.<br>5. The system generates a shareable link for the selected result.<br>6. The system displays the link together with a copy option.<br>7. The user copies the link and shares it outside the app.     |               |
| <b>Sub-flow:</b> S1. The shared page displays the assessment, explanation and evidence only, and does not display the user's profile information or result history.                                                                                                                                                                                                                                             |               |
| <b>Alternative flow:</b> 4a. If the user cancels, the system does not generate a link.<br>5a. If a share link already exists for the result, the system displays the existing link instead of generating a new one.<br>5b. If the link cannot be generated because of a network or database error, the system offers a retry option.<br>7a. If the result is later deleted, the shared link no longer resolves. |               |

## Use Case 16 – User registers for a free account

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User registers for a free account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 16 |
| <b>Stakeholders &amp; Goals:</b> Free user – Gain access to the free functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Description:</b> A new user creates a uiabo free account using an email address and password, and verifies the email address before the account becomes active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Actor(s):</b> Free user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Trigger:</b> The user selects Register on the welcome or login screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is not logged in, has an internet connection, and does not already have an account registered with the same email address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Main Flow:</b> 1. The user opens uiabo and selects Register.<br>2. The system displays the registration form and the available account types.<br>3. The user selects the free account type.<br>4. The user enters their email address and password and confirms the password.<br>5. The system displays the terms of use and privacy notice, and the user gives consent.<br>6. The user submits the registration.<br>7. The system validates the submitted information and checks that the email address is not already registered.<br>8. The system creates the account through Firebase Authentication with the free access tier and an allowance of one successful submission per day, and sends a verification email.<br>9. The user opens the verification link in the registered email inbox.<br>10. The system activates the account, confirms registration and directs the user to log in. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Alternative flow:</b> 7a. If required information is missing or the email format is invalid, the system highlights the relevant field and does not create an account.<br>7b. If the password does not meet the password rules or the two entries do not match, the system explains the requirement and asks the user to correct it.<br>7c. If the email address is already registered, the system informs the user and offers to log in or to reset the password.<br>8a. If the account cannot be created because of a network or service error, the system does not create a partial account and offers a retry option.<br>9a. If the verification link expires before it is used, the system rejects it and offers to send a new verification email.                                                                                                                                             |               |

## Use Case 17 – Free user upgrades to premium

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Free user upgrades to premium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>ID:</b> 17 |
| <b>Stakeholders &amp; Goals:</b> Free user – Access all the functions of the app and gain more uses of the functions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> A logged-in free user subscribes to the premium plan, after which the account tier and submission allowance are updated and the premium functions become available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Actor(s):</b> Free user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Trigger:</b> The user selects Upgrade to Premium from the account menu, the subscription page or a prompt shown when the free allowance is exhausted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Pre-condition:</b> The user is logged in as a free user, does not have an active premium subscription, and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Main Flow:</b> 1. The free user selects Upgrade to Premium.<br>2. The system displays the premium plan, the price, the billing period and the premium allowance and functions.<br>3. The free user confirms the upgrade.<br>4. The system directs the free user to the payment provider.<br>5. The free user enters their payment details on the payment provider's page and authorises the payment.<br>6. The payment provider confirms that the payment was successful.<br>7. The system updates the account tier to premium, applies the premium allowance of up to 60 successful submissions per month and records the subscription expiry date.<br>8. The system confirms the upgrade and makes the premium functions available. |               |
| <b>Sub-flow:</b> S1. Submissions already used on the free allowance are not carried over to the premium allowance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Alternative flow:</b> 3a. If the free user cancels, the system returns to the previous screen and makes no changes.<br>6a. If the payment is declined or cancelled, the system does not upgrade the account and allows the free user to try again.<br>7a. If the payment succeeds but the account cannot be updated, the system does not report the upgrade as complete, records the failure for administrator review and informs the user that support has been notified.<br>8a. If the connection is lost after payment, the system applies the upgrade once the payment confirmation is received.                                                                                                                                  |               |

### 3.2 Use Case Diagram – Premium User

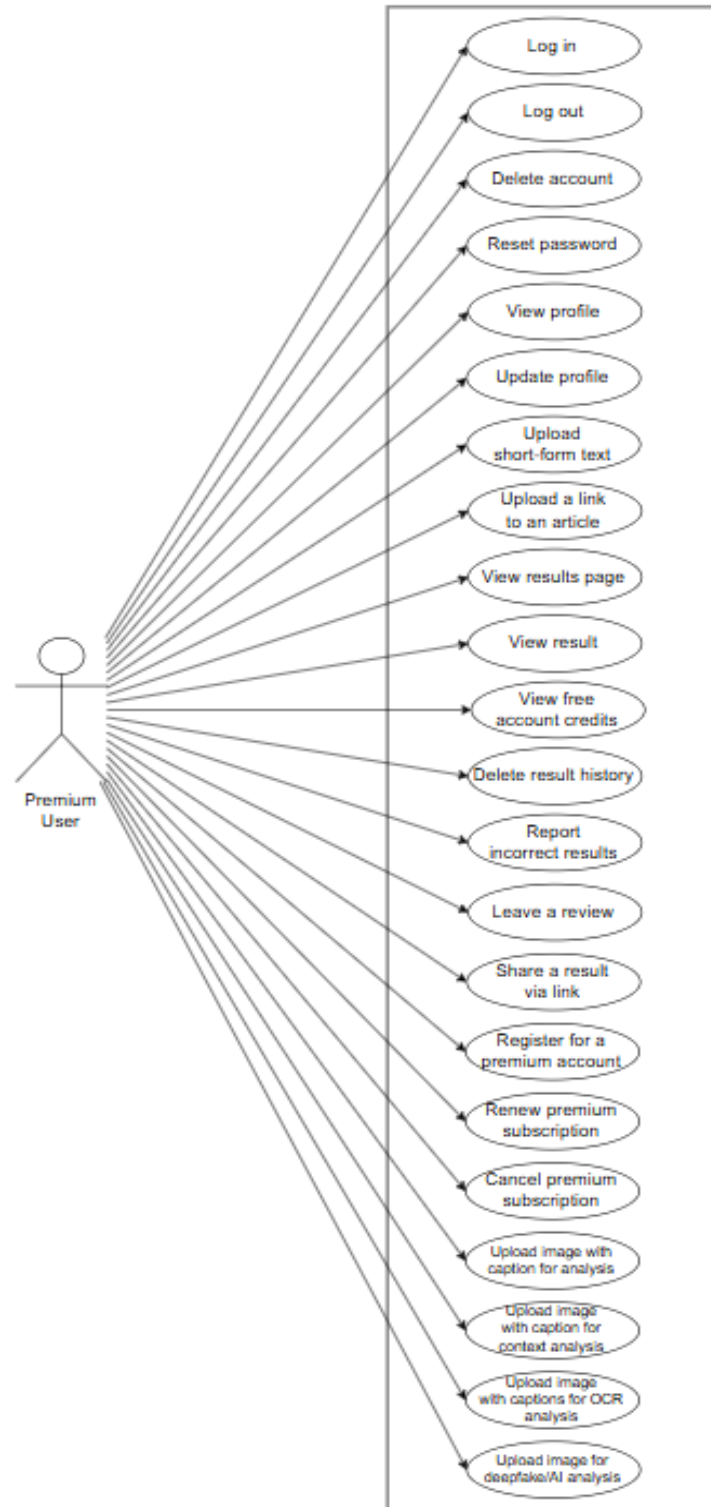


Figure 3.2. Premium User Use Case Diagram

### 3.2.1 User Stories – Premium User

- UC-01: As a premium user, I want to log in to my account so that I can access the functions of the app.
- UC-02: As a premium user, I want to log out of my account so that I can maintain the security of my account.
- UC-03: As a premium user, I want to delete my account so that I can remove my personal data from the system.
- UC-04: As a premium user, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-05: As a premium user, I want to view my profile so that I can check whether my account information is up to date.
- UC-06: As a premium user, I want to update my profile so that I can keep my account information up to date.
- UC-07: As a premium user, I want to upload short-form text so that I can find out whether the submitted text contains misinformation.
- UC-08: As a premium user, I want to upload a link to an article so that I can find out whether a linked article or webpage contains misinformation.
- UC-09: As a premium user, I want to view my personal results page so that I can access the results of my submitted content.
- UC-10: As a premium user, I want to view the result of my submitted content so that I can view the details of a specific analysis result.
- UC-11: As a premium user, I want to view my remaining account credits so that I can find out how many uses of the app I have left.
- UC-12: As a premium user, I want to delete my result history so that I can remove results I no longer want to see from my results page.
- UC-13: As a premium user, I want to report results that are incorrect so that I can provide data and feedback so the system can be improved in future.
- UC-14: As a premium user, I want to leave a review so that I can express my experience using the app so it can be improved in the future.
- UC-15: As a premium user, I want to share a result via link so that I can inform others of content containing misinformation.
- UC-18: As a premium user, I want to register for a premium account so that I can gain access to all the functions of the app.
- UC-19: As a premium user, I want to renew my premium subscription so that I can prevent loss of access to the premium functions of the app.
- UC-20: As a premium user, I want to cancel my premium subscription so that I can stop paying for the premium subscription.
- UC-21: As a premium user, I want to upload an image with a caption for analysis so that I can understand the submitted image and caption and assess potentially misleading content.
- UC-22: As a premium user, I want to upload an image with a caption for context analysis so that I can find out whether the image and caption are contextually consistent.
- UC-23: As a premium user, I want to upload an image for OCR analysis so that I can review visible text extracted from an image and assess it for misinformation.
- UC-24: As a premium user, I want to upload an image for deepfake/AI analysis so that I can assess whether an image may have been generated or manipulated using AI.

### **3.2.2 Use Case Description – Premium User**

Shared Use Cases 01–15 are defined in Section 3.1.2 and apply equally to premium users. The following descriptions define premium registration, subscription and image-analysis behaviour (Use Cases 18–24).

## Use Case 18 – User registers for a premium account

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> User registers for a premium account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ID:</b> 18 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Gain access to all the functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Description:</b> A new user creates a uiabo account and subscribes to the premium plan within a single registration process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Trigger:</b> The user selects Register on the welcome or login screen and selects the premium account type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The user is not logged in, has an internet connection, and does not already have an account registered with the same email address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The user opens uiabo and selects Register.<br>2. The system displays the registration form and the available account types.<br>3. The user selects the premium account type.<br>4. The user enters their email address and password and confirms the password.<br>5. The system displays the terms of use and privacy notice, and the user gives consent.<br>6. The user submits the registration.<br>7. The system validates the submitted information and checks that the email address is not already registered.<br>8. The system directs the user to the payment provider.<br>9. The user enters their payment details on the payment provider's page and authorises the payment.<br>10. The payment provider confirms that the payment was successful.<br>11. The system creates the account through Firebase Authentication with the premium access tier, applies the premium allowance of up to 60 successful submissions per month and sends a verification email.<br>12. The user opens the verification link in the registered email inbox.<br>13. The system activates the account, confirms registration and directs the user to log in. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Alternative flow:</b> 7a. If required information is missing or invalid, the system highlights the relevant field and does not proceed to payment.<br>7b. If the email address is already registered, the system informs the user and offers to log in or to reset the password.<br>10a. If the payment is declined or cancelled, the system does not create an account and offers the user the option to register for a free account instead.<br>11a. If the payment succeeds but the account cannot be created, the system records the failure for administrator review and informs the user that support has been notified.<br>12a. If the verification link expires before it is used, the system rejects it and offers to send a new verification email.                                                                                                                                                                                                                                                                                                                                                                                                           |               |

## Use Case 19 – Premium user renews their premium subscription

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user renews their premium subscription                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>ID:</b> 19 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Prevent loss of access to the premium functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Description:</b> A premium user extends an expiring or expired subscription for a further billing period so that premium access and the premium allowance continue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Trigger:</b> The premium user selects Renew Subscription on the subscription page, or acts on the renewal reminder sent by the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Pre-condition:</b> The user is logged in as a premium user, has a subscription that is due to expire or has expired, and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Main Flow:</b> 1. The system notifies the premium user before the subscription expiry date.<br>2. The premium user opens the subscription page.<br>3. The premium user selects Renew Subscription.<br>4. The system displays the renewal price, the billing period and the new expiry date.<br>5. The premium user confirms the renewal.<br>6. The system directs the premium user to the payment provider.<br>7. The premium user enters their payment details on the payment provider's page and authorises the payment.<br>8. The payment provider confirms that the payment was successful.<br>9. The system extends the subscription, updates the expiry date and refreshes the premium allowance for the new period.<br>10. The system confirms the renewal and displays the new expiry date. |               |
| <b>Sub-flow:</b> S1. If automatic renewal is enabled, the system attempts the renewal payment on the expiry date without further action from the premium user.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Alternative flow:</b> 5a. If the premium user cancels, the system makes no changes and the existing expiry date remains.<br>8a. If the payment is declined or cancelled, the system does not extend the subscription and informs the premium user when premium access will end.<br>9a. If the subscription has already been renewed, the system informs the premium user and does not take a further payment.<br>10a. If the subscription expired before it was renewed, the system restores premium access and applies the premium allowance from the renewal date.                                                                                                                                                                                                                                |               |

## Use Case 20 – Premium user cancels their premium subscription

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user cancels their premium subscription                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>ID:</b> 20 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Stop paying for the premium subscription.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Description:</b> A premium user cancels the automatic renewal of the premium subscription. Premium access continues until the current expiry date, after which the account reverts to the free tier.                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The premium user selects Cancel Subscription on the subscription page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Pre-condition:</b> The user is logged in as a premium user and has an active premium subscription.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Main Flow:</b> 1. The premium user opens the subscription page.<br>2. The premium user selects Cancel Subscription.<br>3. The system explains that premium access continues until the current expiry date and that the account will revert to the free tier afterwards.<br>4. The premium user confirms the cancellation.<br>5. The system cancels the automatic renewal and updates the subscription status to cancelled.<br>6. The system confirms the cancellation and displays the date on which premium access will end.<br>7. The system sends a confirmation to the registered email address. |               |
| <b>Sub-flow:</b> S1. On the expiry date the system reverts the account to the free tier and applies the free allowance of one successful submission per day.<br>S2. Saved results remain available to the user after the account reverts to the free tier.                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Alternative flow:</b> 4a. If the premium user cancels the confirmation, the subscription remains unchanged.<br>5a. If the cancellation cannot be saved because of a network or database error, the system does not confirm the cancellation and offers a retry option.<br>6a. Cancelling the subscription does not delete the account. The user deletes the account through the separate account deletion function.                                                                                                                                                                                  |               |

## Use Case 21 – Premium user uploads image with captions for analysis

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user uploads image with captions for analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ID:</b> 21 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Understand the submitted image and caption and assess potentially misleading content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Description:</b> A premium user uploads a supported static image with an accompanying English caption for combined content analysis and evidence-backed result generation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Trigger:</b> The premium user selects Image and Caption Analysis from the submission screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Pre-condition:</b> The user is logged in with active premium access, has a remaining monthly submission allowance, has an internet connection and has a supported image they are permitted to process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Main Flow:</b> 1. The user selects Image and Caption Analysis.<br>2. The system verifies premium entitlement and displays the upload form and remaining monthly allowance.<br>3. The user selects a supported static image.<br>4. The system displays a preview and a field for an accompanying caption.<br>5. The user enters or edits the English caption and selects Submit.<br>6. The system validates the image, caption and allowance, then uploads the submission securely.<br>7. The back end analyses the image and caption, identifies checkable content and retrieves relevant evidence.<br>8. When processing succeeds, the system records the result, deducts one monthly submission and displays the result page with explanation, evidence, uncertainty and next-action guidance. |               |
| <b>Sub-flow:</b> S1. The user may replace the selected image or edit the caption before submitting.<br>S2. From the preview, the user may choose the specialised context, OCR or image-authenticity function and continue with Use Case 22, 23 or 24.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Alternative flow:</b> 2a. If premium access is inactive, the system blocks the function and shows subscription guidance.<br>2b. If no monthly allowance remains, the system displays the next refresh information and does not accept the submission.<br>6a. If the image is missing, unsupported, corrupted or exceeds the permitted size, the system explains the problem and requests another image.<br>6b. If the caption is missing or cannot be processed as English text, the system asks the user to correct it.<br>7a. If processing fails, is cancelled or exceeds 20 minutes, the system records no successful use, preserves the allowance and offers a safe retry.<br>8a. If evidence is insufficient, the result uses Not Enough Information and explains the limitation.          |               |

## Use Case 22 – Premium user uploads image with captions for context analysis

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user uploads image with captions for context analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ID:</b> 22 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Find out whether the image and caption are contextually consistent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Description:</b> A premium user submits a static image and caption so the system can compare contextual cues such as date, location, event and description and identify a possible mismatch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Trigger:</b> The premium user selects Caption-Image Context Analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Pre-condition:</b> The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image and English caption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Main Flow:</b> 1. The user selects Caption-Image Context Analysis.<br>2. The system verifies premium entitlement and displays the upload form and allowance.<br>3. The user selects an image, enters or edits its caption and reviews the preview.<br>4. The user selects Analyse Context.<br>5. The system validates and securely uploads the submission.<br>6. The back end identifies contextual cues in the image and caption and searches for relevant corroborating evidence.<br>7. The system compares the available cues and generates a context-assessment result with uncertainty and limitations.<br>8. On success, the system records the result, deducts one monthly submission and displays the explanation, evidence links and recommended next action. |               |
| <b>Sub-flow:</b> S1. If useful visible text is detected, the user may review OCR-extracted text before analysis by continuing with Use Case 23.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Alternative flow:</b> 2a. If premium access is inactive or no allowance remains, the system does not accept the submission and explains the available action.<br>5a. If an input is missing, unsupported or invalid, the system highlights the issue and does not start processing.<br>6a. If no reliable contextual evidence can be found, the system returns Not Enough Information rather than asserting a mismatch.<br>7a. If analysis fails, is cancelled or times out, the allowance is not deducted and the system offers a retry option.                                                                                                                                                                                                                       |               |

### Use Case 23 – Premium user uploads image with captions for OCR analysis

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user uploads image with captions for OCR analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ID:</b> 23 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Review visible text extracted from an image and assess it for misinformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Description:</b> A premium user uploads an image containing visible text, reviews and corrects the OCR output, and submits the corrected text for evidence-backed misinformation analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Trigger:</b> The premium user selects OCR Image Analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Pre-condition:</b> The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image containing legible text.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The user selects OCR Image Analysis.<br>2. The system verifies premium entitlement and displays the image-upload form and allowance.<br>3. The user selects an image and chooses Extract Text.<br>4. The system validates the image, uploads it securely and performs OCR.<br>5. The system displays the extracted text in an editable field beside the image preview.<br>6. The user checks the extracted text, corrects any errors and selects Analyse Text.<br>7. The system validates the corrected text, identifies checkable claims and retrieves relevant evidence.<br>8. On successful analysis, the system records the result, deducts one monthly submission and displays the concern label, risk, uncertainty, explanation and citations. |               |
| <b>Sub-flow:</b> S1. The user may re-run OCR after replacing or rotating the image before submitting the corrected text.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Alternative flow:</b> 2a. If premium access is inactive or no allowance remains, the system blocks the analysis and explains why.<br>4a. If the image is unsupported, corrupted or too unclear for usable OCR, the system requests a clearer supported image and does not deduct an allowance.<br>6a. If the corrected text is empty or contains no analysable content, the system requests a correction and does not start misinformation analysis.<br>7a. If no objectively checkable claim is identified, the system explains this and returns Not Enough Information where appropriate.<br>8a. If processing fails, is cancelled or times out, the system preserves the allowance and offers a retry option.                                                       |               |

## Use Case 24 – Premium user uploads image for deepfake/AI analysis

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Premium user uploads image for deepfake/AI analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ID:</b> 24 |
| <b>Stakeholders &amp; Goals:</b> Premium user – Assess whether an image may have been generated or manipulated using AI.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Description:</b> A premium user submits a supported static image to receive a probabilistic AI-generation or manipulation likelihood score and a plain-language media-forensics explanation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Actor(s):</b> Premium user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Trigger:</b> The premium user selects Deepfake/AI Image Analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Pre-condition:</b> The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image they are permitted to process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The user selects Deepfake/AI Image Analysis.<br>2. The system verifies premium entitlement and displays the upload form, limitations notice and remaining allowance.<br>3. The user selects an image and reviews the preview.<br>4. The user acknowledges that the output is a likelihood assessment rather than proof and selects Analyse Image.<br>5. The system validates and securely uploads the image.<br>6. The back end runs the supported image-authenticity detector and gathers the available signals.<br>7. The system converts the detector output into a likelihood score, uncertainty statement and plain-language explanation of the available signals and limitations, including suspicious-region indicators only when supported.<br>8. On success, the system records the result, deducts one monthly submission and displays the result and safe next-action guidance. |               |
| <b>Sub-flow:</b> S1. The user may add an optional caption to provide context, but the image-authenticity score remains distinct from the misinformation-risk assessment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Alternative flow:</b> 2a. If premium access is inactive or no allowance remains, the system blocks the function and explains the available action.<br>5a. If the image is unsupported, corrupted or too low-quality for the detector, the system asks for another image and does not deduct an allowance.<br>6a. If the detector is unavailable or cannot produce a reliable output, the system returns Not Enough Information rather than a conclusive label.<br>8a. If processing fails, is cancelled or exceeds 20 minutes, the system preserves the allowance and offers a retry option.                                                                                                                                                                                                                                                                                                                 |               |

### 3.3 Use Case Diagram – System Administrator

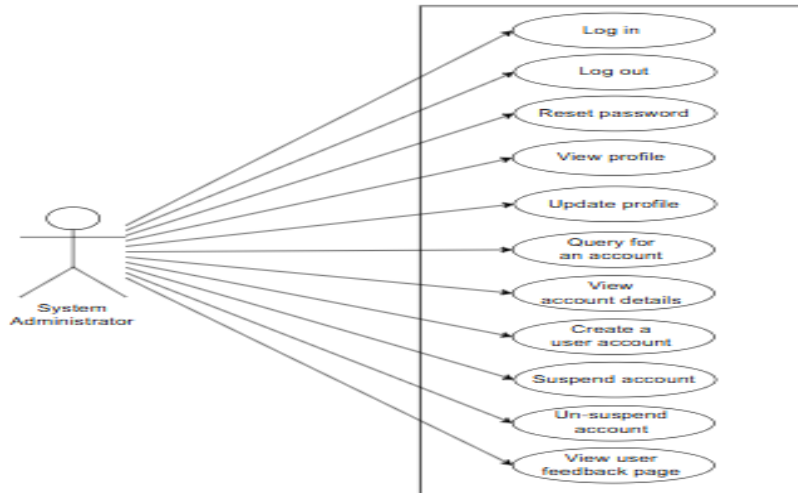


Figure 3.3. System Administrator Use Case Diagram

### **3.3.1 User Stories – System Administrator**

- UC-25: As a system administrator, I want to log in to my account so that I can access the administrative functions of the app.
- UC-26: As a system administrator, I want to log out of my account so that I can maintain the security of my account.
- UC-27: As a system administrator, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-28: As a system administrator, I want to view my profile so that I can check whether my information is up to date.
- UC-29: As a system administrator, I want to update my profile so that I can keep my information up to date.
- UC-30: As a system administrator, I want to query for an account so that I can find a specific account quickly.
- UC-31: As a system administrator, I want to view account details so that I can verify the account's information and access status.
- UC-32: As a system administrator, I want to create a user account so that I can create an account for a new system administrator or data engineer.
- UC-33: As a system administrator, I want to suspend an account so that I can revoke a specific account's access without deleting its record.
- UC-34: As a system administrator, I want to un-suspend an account so that I can restore a specific suspended account's access.
- UC-35: As a system administrator, I want to view the user feedback page so that I can review feedback and reviews provided by users.

### **3.3.2 Use Case Description – System Administrator**

## Use Case 25 – System administrator logs in to their account

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| <b>Name:</b> System administrator logs in to their account                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 25 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Access the administrative functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Description:</b> An authorised system administrator authenticates to access role-restricted uiabo functions and information.                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Trigger:</b> The system administrator selects Log In on the login screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Pre-condition:</b> The system administrator has an active account created by an authorised system administrator, is not already logged in and has an internet connection.                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Main Flow:</b> 1. The system administrator opens uiabo and selects Log In.<br>2. The system displays the login form.<br>3. The system administrator enters their registered email address and password.<br>4. The system administrator selects Log In.<br>5. The system validates the credentials through Firebase Authentication.<br>6. The system verifies that the account is active and has the system administrator role.<br>7. The system creates the authenticated session and displays the administrative home screen. |               |
| <b>Sub-flow:</b> S1. If the system administrator has forgotten the password, they select Forgot Password and continue with Use Case 27.                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Alternative flow:</b> 5a. If the credentials are invalid, the system denies access and displays a clear error message.<br>6a. If the account is suspended or has the wrong role, the system denies access without exposing protected information.<br>7a. If a network or service error occurs, the system keeps the user logged out and offers a retry option.                                                                                                                                                                 |               |

**Use Case 26 – System administrator logs out of their account**

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator logs out of their account                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 26 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Maintain the security of their account.                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> An authenticated system administrator ends the current session so that protected operational functions are no longer accessible from the device.                                                                                                                                                                                                                               |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Trigger:</b> The system administrator selects Log Out from the account or profile menu.                                                                                                                                                                                                                                                                                                         |               |
| <b>Pre-condition:</b> The system administrator is logged in to an active session.                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The system administrator opens the account or profile menu.<br>2. The user selects Log Out.<br>3. The system asks for confirmation.<br>4. The user confirms the action.<br>5. The system ends the authenticated session and clears locally stored session credentials.<br>6. The system returns to the login screen and no protected operational information remains visible. |               |
| <b>Sub-flow:</b> S1. After 30 minutes of inactivity, the system automatically expires the session and returns to the login screen.                                                                                                                                                                                                                                                                 |               |
| <b>Alternative flow:</b> 4a. If the user cancels, the system keeps the session active.<br>5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and reports that the service could not be reached.                                                                                                                                            |               |

## Use Case 27 – System administrator resets their password

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator resets their password                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ID:</b> 27 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Improve account security or regain access when the password is forgotten.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Description:</b> The system administrator requests a secure password-reset link and replaces the existing password with a valid new password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Trigger:</b> The system administrator selects Forgot Password on the login screen or Reset Password in profile settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The system administrator has a registered email-and-password account, can access the registered inbox and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Main Flow:</b> 1. The system administrator selects Forgot Password or Reset Password.<br>2. The system displays the password-reset form.<br>3. The user enters the registered email address and submits the request.<br>4. The system sends a time-limited password-reset link and displays a neutral confirmation.<br>5. The user opens the link and enters the new password twice.<br>6. The system validates the link, password match and password rules.<br>7. Firebase Authentication updates the password and invalidates the link.<br>8. The system confirms the change and directs the user to log in again. |               |
| <b>Sub-flow:</b> S1. A logged-in user may start the same process from profile settings as a preventative security measure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Alternative flow:</b> 3a. If the email format is invalid, the system highlights the field and does not submit the request.<br>4a. The system uses the same neutral confirmation whether or not an account can be disclosed.<br>6a. If the link is invalid, expired or used, the system rejects it and offers to send another link.<br>6b. If the passwords do not match or fail the password rules, the system explains the problem and requests a correction.<br>7a. If the network or authentication service is unavailable, the password remains unchanged and the system offers a retry option.                  |               |

**Use Case 28 – System administrator views their profile**

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator views their profile                                                                                                                                                                                                                                                                               | <b>ID:</b> 28 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Check whether their information is up to date.                                                                                                                                                                                                                              |               |
| <b>Description:</b> The system administrator views the profile information stored for their own operational account.                                                                                                                                                                                                                |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                               |               |
| <b>Trigger:</b> The system administrator selects Profile from the navigation or account menu.                                                                                                                                                                                                                                       |               |
| <b>Pre-condition:</b> The system administrator is logged in to an active account.                                                                                                                                                                                                                                                   |               |
| <b>Main Flow:</b> 1. The user selects Profile.<br>2. The system verifies the current session and role.<br>3. The system retrieves only the authenticated user's profile information.<br>4. The system displays the profile information and operational role in a readable format.<br>5. The user reviews the displayed information. |               |
| <b>Sub-flow:</b> S1. The user may select Edit Profile and continue with Use Case 29.                                                                                                                                                                                                                                                |               |
| <b>Alternative flow:</b> 2a. If the session has expired, the system returns to the login screen.<br>3a. If the profile cannot be retrieved, the system displays an error without exposing another account's information and offers a retry option.                                                                                  |               |

## Use Case 29 – System administrator updates their profile

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator updates their profile                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ID:</b> 29 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Keep their information up to date.                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Description:</b> The system administrator changes editable information in their own profile while protected role and access-control fields remain read-only.                                                                                                                                                                                                                                                                                                        |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Trigger:</b> The system administrator selects Edit Profile while viewing their profile.                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The system administrator is logged in and the existing profile information is available.                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Main Flow:</b> 1. The user views their profile and selects Edit Profile.<br>2. The system displays editable fields with their current values; the role is read-only.<br>3. The user changes one or more editable fields and selects Save Changes.<br>4. The system validates the information.<br>5. The system saves the valid changes to the authenticated user's profile.<br>6. The system displays the updated profile and confirms that the changes were saved. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Alternative flow:</b> 3a. If the user cancels, the system discards unsaved changes.<br>4a. If required information is missing or invalid, the system highlights the relevant field and does not save the changes.<br>5a. If an authentication, network or database error occurs, the previous profile remains unchanged and the system offers a retry option.                                                                                                       |               |

### Use Case 30 – System administrator queries for an account

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| <b>Name:</b> System administrator queries for an account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ID:</b> 30 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Find a specific account quickly.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Description:</b> A system administrator searches authorised account records using one or more supported identifiers or filters.                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Trigger:</b> The system administrator selects Account Management or uses the account search field.                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Pre-condition:</b> The system administrator is logged in with an active administrative account.                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Main Flow:</b> 1. The administrator opens Account Management.<br>2. The system displays the account search and permitted filters.<br>3. The administrator enters a supported query such as account email, identifier, role, tier or status.<br>4. The administrator selects Search.<br>5. The system validates the query and applies role-based access controls.<br>6. The system displays matching accounts with only the summary fields required to distinguish them.<br>7. The system records the administrative query in the audit log. |               |
| <b>Sub-flow:</b> S1. The administrator may refine, sort or clear the query and search again.<br>S2. The administrator may select a result and continue with Use Case 31.                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Alternative flow:</b> 3a. If the query is empty, malformed or too broad, the system requests a more specific valid value.<br>6a. If no account matches, the system displays No Results without suggesting protected account information.<br>6b. If the database cannot be reached, the system displays an error and offers a retry option.                                                                                                                                                                                                  |               |

### Use Case 31 – System administrator views account details

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator views account details                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ID:</b> 31 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Verify the account's information and access status.                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Description:</b> A system administrator views the authorised details, role, tier and status of a selected account before performing an account-management action.                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Trigger:</b> The system administrator selects an account from the query results.                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Pre-condition:</b> The administrator is logged in and the selected account is within the administrator's permitted scope.                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Main Flow:</b> 1. The administrator queries for an account using Use Case 30.<br>2. The administrator selects a matching account.<br>3. The system verifies the session, administrative role and access scope.<br>4. The system retrieves the account's permitted profile summary, role, access tier, status and relevant timestamps.<br>5. The system displays the details while masking or excluding sensitive fields that are not required for administration.<br>6. The system records the view in the audit log. |               |
| <b>Sub-flow:</b> S1. From the account-details screen, the administrator may start Use Case 33 or 34 when the selected account is eligible.                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Alternative flow:</b> 3a. If the session has expired or access is not permitted, the system denies the request and displays no account details.<br>4a. If the account no longer exists or cannot be retrieved, the system displays an error and returns to the search results.                                                                                                                                                                                                                                        |               |

## Use Case 32 – System administrator creates a user account

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator creates a user account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ID:</b> 32 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Create an account for a new system administrator or data engineer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Description:</b> An authorised system administrator creates an operational account with the appropriate system-administrator or data-engineer role.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Trigger:</b> The system administrator selects Create Operational Account.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The administrator is logged in, has permission to create operational accounts and has the new user's required information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Main Flow:</b> 1. The administrator opens Account Management and selects Create Operational Account.<br>2. The system displays the authorised account fields and role choices.<br>3. The administrator enters the required information and selects System Administrator or Data Engineer.<br>4. The administrator reviews the details and confirms creation.<br>5. The system validates the data, the administrator's permission and the uniqueness of the account identifier.<br>6. The system creates the authentication account and associated role/profile record in a consistent transaction or recoverable workflow.<br>7. The system sends the new user a secure account-activation or password-setup instruction.<br>8. The system confirms creation and records the action in the audit log. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Alternative flow:</b> 4a. If the administrator cancels, the system creates no account.<br>5a. If required data is missing, invalid or already registered, the system highlights the problem and does not create the account.<br>5b. If the selected role is outside the administrator's authority, the system denies the request.<br>6a. If any required operation fails, the system does not report success, safely rolls back or flags incomplete creation and offers a retry or support path.                                                                                                                                                                                                                                                                                                      |               |

### Use Case 33 – System administrator suspends account

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator suspends account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ID:</b> 33 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Revoke a specific account's access without deleting its record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Description:</b> A system administrator changes an eligible active account to suspended status so that it cannot authenticate or use protected functions while its record remains available for authorised administration.                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Trigger:</b> The system administrator selects Suspend Account on an eligible account-details screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Pre-condition:</b> The administrator is logged in, has suspension permission and has selected an active account that is eligible for suspension.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Main Flow:</b> 1. The administrator views the selected account using Use Case 31.<br>2. The administrator selects Suspend Account.<br>3. The system explains the effect, requests a reason and asks for confirmation.<br>4. The administrator enters the reason and confirms.<br>5. The system rechecks authority and safeguards against suspending the current session or removing the last active system administrator.<br>6. The system changes the account status to Suspended and revokes active sessions where supported.<br>7. The system confirms the action and records the administrator, account, reason and timestamp in the audit log. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Alternative flow:</b> 4a. If the administrator cancels or provides no required reason, the account remains active.<br>5a. If the target is the current administrator or the last active system administrator, the system blocks the action and explains the safeguard.<br>6a. If the status update fails, the system leaves the account's previous status unchanged and offers a retry option.                                                                                                                                                                                                                                                      |               |

### Use Case 34 – System administrator un-suspends account

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator un-suspends account                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ID:</b> 34 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Restore a specific suspended account's access.                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Description:</b> A system administrator restores an eligible suspended account to active status without creating a new account record.                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Trigger:</b> The system administrator selects Unsuspend Account on a suspended account-details screen.                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Pre-condition:</b> The administrator is logged in, has unsuspension permission and has selected a suspended account.                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Main Flow:</b> 1. The administrator views the suspended account using Use Case 31.<br>2. The administrator selects Unsuspend Account.<br>3. The system displays the current suspension information and asks for confirmation.<br>4. The administrator confirms the action.<br>5. The system verifies the administrator's authority and the target account's current status.<br>6. The system changes the account status to Active.<br>7. The system confirms the action and records it in the audit log. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Alternative flow:</b> 4a. If the administrator cancels, the account remains suspended.<br>5a. If the account is already active or no longer exists, the system does not apply a duplicate update and refreshes the details.<br>6a. If the update fails, the account remains suspended and the system offers a retry option.                                                                                                                                                                              |               |

### Use Case 35 – System administrator views user feedback page

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> System administrator views user feedback page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 35 |
| <b>Stakeholders &amp; Goals:</b> System administrator – Review feedback and reviews provided by users.                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| <b>Description:</b> A system administrator views and filters user reviews, feedback and reported service problems so recurring issues can be identified and followed up.                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Actor(s):</b> System administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Trigger:</b> The system administrator selects User Feedback from the administrative navigation.                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Pre-condition:</b> The administrator is logged in with permission to review feedback.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Main Flow:</b> 1. The administrator selects User Feedback.<br>2. The system verifies the session and administrative role.<br>3. The system retrieves the feedback records within the administrator's permitted scope.<br>4. The system displays a summary and a list containing the feedback category, status, submission date and only the user information required for follow-up.<br>5. The administrator filters, sorts or selects an item.<br>6. The system displays the selected feedback details and records the administrative access. |               |
| <b>Sub-flow:</b> S1. A reported incorrect analysis may link to the data-engineering workflow in Use Cases 41 and 42 while access remains role-controlled.                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Alternative flow:</b> 3a. If no feedback is available, the system displays an empty state.<br>3b. If feedback cannot be retrieved, the system displays an error and offers a retry option.<br>6a. If the item was removed or access is no longer permitted, the system displays no details and refreshes the list.                                                                                                                                                                                                                             |               |

### 3.4 Use Case Diagram – Data Engineer

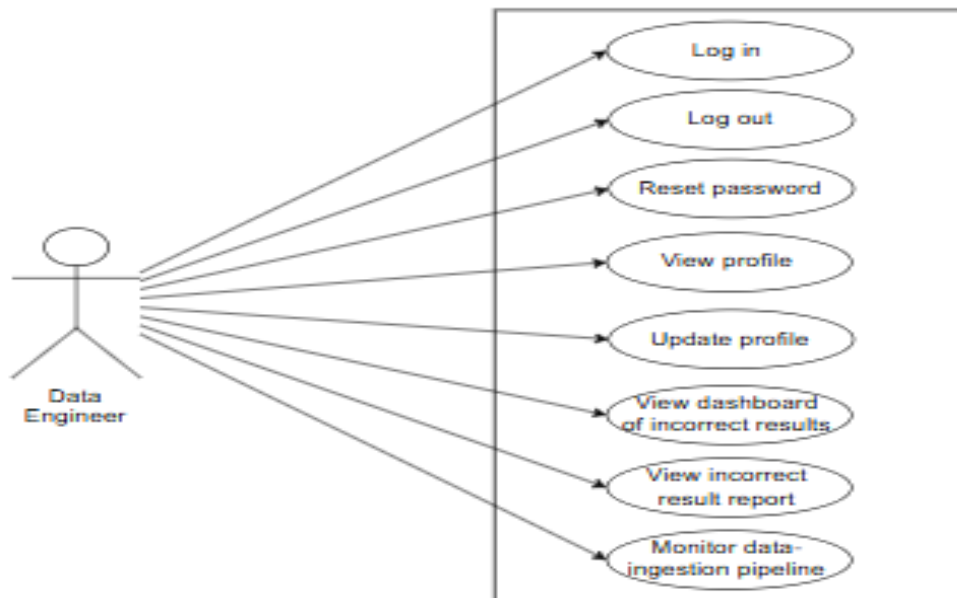


Figure 3.4. Data Engineer Use Case Diagram

### **3.4.1 User Stories – Data Engineer**

- UC-36: As a data engineer, I want to log in to my account so that I can access the data-engineering functions of the app.
- UC-37: As a data engineer, I want to log out of my account so that I can maintain the security of my account.
- UC-38: As a data engineer, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-39: As a data engineer, I want to view my profile so that I can check whether my information is up to date.
- UC-40: As a data engineer, I want to update my profile so that I can keep my information up to date.
- UC-41: As a data engineer, I want to view the dashboard of incorrect results so that I can analyse reported incorrect results and identify possible flaws in the AI models or data workflow.
- UC-42: As a data engineer, I want to view incorrect-result reports so that I can view the details of a reported incorrect result for investigation.
- UC-43: As a data engineer, I want to monitor data-ingestion pipelines so that I can identify failed, delayed or incomplete data updates.

### **3.4.2 Use Case Description – Data Engineer**

### Use Case 36 – Data engineer logs in to their account

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer logs in to their account                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ID:</b> 36 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Access the data-engineering functions of the app.                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> An authorised data engineer authenticates to access role-restricted uiabo functions and information.                                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Trigger:</b> The data engineer selects Log In on the login screen.                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Pre-condition:</b> The data engineer has an active account created by an authorised system administrator, is not already logged in and has an internet connection.                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Main Flow:</b> 1. The data engineer opens uiabo and selects Log In.<br>2. The system displays the login form.<br>3. The data engineer enters their registered email address and password.<br>4. The data engineer selects Log In.<br>5. The system validates the credentials through Firebase Authentication.<br>6. The system verifies that the account is active and has the data engineer role.<br>7. The system creates the authenticated session and displays the data-engineering dashboard. |               |
| <b>Sub-flow:</b> S1. If the data engineer has forgotten the password, they select Forgot Password and continue with Use Case 38.                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Alternative flow:</b> 5a. If the credentials are invalid, the system denies access and displays a clear error message.<br>6a. If the account is suspended or has the wrong role, the system denies access without exposing protected information.<br>7a. If a network or service error occurs, the system keeps the user logged out and offers a retry option.                                                                                                                                     |               |

### Use Case 37 – Data engineer logs out of their account

|                                                                                                                                                                                                                                                                                                                                                                                             |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer logs out of their account                                                                                                                                                                                                                                                                                                                                        | <b>ID:</b> 37 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Maintain the security of their account.                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> An authenticated data engineer ends the current session so that protected operational functions are no longer accessible from the device.                                                                                                                                                                                                                               |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                              |               |
| <b>Trigger:</b> The data engineer selects Log Out from the account or profile menu.                                                                                                                                                                                                                                                                                                         |               |
| <b>Pre-condition:</b> The data engineer is logged in to an active session.                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The data engineer opens the account or profile menu.<br>2. The user selects Log Out.<br>3. The system asks for confirmation.<br>4. The user confirms the action.<br>5. The system ends the authenticated session and clears locally stored session credentials.<br>6. The system returns to the login screen and no protected operational information remains visible. |               |
| <b>Sub-flow:</b> S1. After 30 minutes of inactivity, the system automatically expires the session and returns to the login screen.                                                                                                                                                                                                                                                          |               |
| <b>Alternative flow:</b> 4a. If the user cancels, the system keeps the session active.<br>5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and reports that the service could not be reached.                                                                                                                                     |               |

### Use Case 38 – Data engineer resets their password

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer resets their password                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ID:</b> 38 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Improve account security or regain access when the password is forgotten.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Description:</b> The data engineer requests a secure password-reset link and replaces the existing password with a valid new password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <b>Trigger:</b> The data engineer selects Forgot Password on the login screen or Reset Password in profile settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Pre-condition:</b> The data engineer has a registered email-and-password account, can access the registered inbox and has an internet connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Main Flow:</b> 1. The data engineer selects Forgot Password or Reset Password.<br>2. The system displays the password-reset form.<br>3. The user enters the registered email address and submits the request.<br>4. The system sends a time-limited password-reset link and displays a neutral confirmation.<br>5. The user opens the link and enters the new password twice.<br>6. The system validates the link, password match and password rules.<br>7. Firebase Authentication updates the password and invalidates the link.<br>8. The system confirms the change and directs the user to log in again. |               |
| <b>Sub-flow:</b> S1. A logged-in user may start the same process from profile settings as a preventative security measure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <b>Alternative flow:</b> 3a. If the email format is invalid, the system highlights the field and does not submit the request.<br>4a. The system uses the same neutral confirmation whether or not an account can be disclosed.<br>6a. If the link is invalid, expired or used, the system rejects it and offers to send another link.<br>6b. If the passwords do not match or fail the password rules, the system explains the problem and requests a correction.<br>7a. If the network or authentication service is unavailable, the password remains unchanged and the system offers a retry option.           |               |

**Use Case 39 – Data engineer views their profile**

|                                                                                                                                                                                                                                                                                                                                     |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer views their profile                                                                                                                                                                                                                                                                                      | <b>ID:</b> 39 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Check whether their information is up to date.                                                                                                                                                                                                                                     |               |
| <b>Description:</b> The data engineer views the profile information stored for their own operational account.                                                                                                                                                                                                                       |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                      |               |
| <b>Trigger:</b> The data engineer selects Profile from the navigation or account menu.                                                                                                                                                                                                                                              |               |
| <b>Pre-condition:</b> The data engineer is logged in to an active account.                                                                                                                                                                                                                                                          |               |
| <b>Main Flow:</b> 1. The user selects Profile.<br>2. The system verifies the current session and role.<br>3. The system retrieves only the authenticated user's profile information.<br>4. The system displays the profile information and operational role in a readable format.<br>5. The user reviews the displayed information. |               |
| <b>Sub-flow:</b> S1. The user may select Edit Profile and continue with Use Case 40.                                                                                                                                                                                                                                                |               |
| <b>Alternative flow:</b> 2a. If the session has expired, the system returns to the login screen.<br>3a. If the profile cannot be retrieved, the system displays an error without exposing another account's information and offers a retry option.                                                                                  |               |

### Use Case 40 – Data engineer updates their profile

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer updates their profile                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ID:</b> 40 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Keep their information up to date.                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> The data engineer changes editable information in their own profile while protected role and access-control fields remain read-only.                                                                                                                                                                                                                                                                                                               |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Trigger:</b> The data engineer selects Edit Profile while viewing their profile.                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Pre-condition:</b> The data engineer is logged in and the existing profile information is available.                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Main Flow:</b> 1. The user views their profile and selects Edit Profile.<br>2. The system displays editable fields with their current values; the role is read-only.<br>3. The user changes one or more editable fields and selects Save Changes.<br>4. The system validates the information.<br>5. The system saves the valid changes to the authenticated user's profile.<br>6. The system displays the updated profile and confirms that the changes were saved. |               |
| <b>Sub-flow:</b> None.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Alternative flow:</b> 3a. If the user cancels, the system discards unsaved changes.<br>4a. If required information is missing or invalid, the system highlights the relevant field and does not save the changes.<br>5a. If an authentication, network or database error occurs, the previous profile remains unchanged and the system offers a retry option.                                                                                                       |               |

## Use Case 41 – Data engineer views dashboard of incorrect results

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer views dashboard of incorrect results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>ID:</b> 41 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Analyse reported incorrect results and identify possible flaws in the AI models or data workflow.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Description:</b> A data engineer views aggregated information about user-reported incorrect results, including counts, trends and supported categories, to prioritise investigation.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
| <b>Trigger:</b> The data engineer selects Incorrect Results Dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| <b>Pre-condition:</b> The data engineer is logged in with an active data-engineer account and authorised report data is available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Main Flow:</b> 1. The data engineer selects Incorrect Results Dashboard.<br>2. The system verifies the session and data-engineer role.<br>3. The system retrieves authorised report and evaluation summary data.<br>4. The system displays dashboard metrics and trends such as report volume, input type, concern label, model or pipeline component and report status where those fields are available.<br>5. The data engineer changes the time range or applies supported filters.<br>6. The system refreshes the dashboard using the selected criteria.<br>7. The data engineer selects a report or group requiring investigation. |               |
| <b>Sub-flow:</b> S1. The data engineer may continue from a selected report to Use Case 42.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Alternative flow:</b> 3a. If no reports match, the system displays an empty state rather than misleading zeroed charts.<br>3b. If data is incomplete, the system identifies the affected metric or time range.<br>4a. If the dashboard cannot be generated, the system displays an error and offers a retry option without exposing partial protected data.                                                                                                                                                                                                                                                                             |               |

## Use Case 42 – Data engineer views incorrect results reports

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer views incorrect results reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ID:</b> 42 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – View the details of a reported incorrect result for investigation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> A data engineer inspects a selected user report together with the associated submission, generated result, model metadata, evidence and provenance information that the role is authorised to view.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| <b>Trigger:</b> The data engineer selects an incorrect-result report from the dashboard or report list.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |
| <b>Pre-condition:</b> The data engineer is logged in, the report exists and the associated data is within the role's permitted scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <b>Main Flow:</b> 1. The data engineer opens a report from the dashboard or report list.<br>2. The system verifies the session, role and access scope.<br>3. The system retrieves the user's report reason, relevant submitted content, generated assessment, evidence references, timestamps and available model or pipeline metadata.<br>4. The system displays the information in sections and masks personal information that is not required for investigation.<br>5. The data engineer reviews the discrepancy and supporting provenance.<br>6. The data engineer records an investigation note or permitted status update.<br>7. The system validates and saves the update and records it in the audit history. |               |
| <b>Sub-flow:</b> S1. The data engineer may open a related pipeline record and continue with Use Case 43.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <b>Alternative flow:</b> 2a. If access is not permitted or the session has expired, the system displays no report information.<br>3a. If the report or associated result is unavailable, the system identifies the missing record and returns to the list.<br>6a. If the user cancels an edit, the system discards the unsaved note or status change.<br>7a. If saving fails, the previous report state remains unchanged and the system offers a retry option.                                                                                                                                                                                                                                                        |               |

### Use Case 43 – Data engineer monitors data-ingestion pipelines

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Name:</b> Data engineer monitors data-ingestion pipelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ID:</b> 43 |
| <b>Stakeholders &amp; Goals:</b> Data engineer – Identify failed, delayed or incomplete data updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Description:</b> A data engineer monitors authorised data-ingestion pipelines, source provenance and run status so failures or incomplete updates that may affect the verification workflow can be investigated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Actor(s):</b> Data engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <b>Trigger:</b> The data engineer selects Data-Ingestion Pipelines from the data-engineering navigation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
| <b>Pre-condition:</b> The data engineer is logged in and pipeline monitoring data is available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <b>Main Flow:</b> 1. The data engineer selects Data-Ingestion Pipelines.<br>2. The system verifies the session and data-engineer role.<br>3. The system retrieves the latest authorised pipeline runs, status, source, timestamps and data-quality indicators.<br>4. The system displays pipeline health and clearly marks successful, running, delayed, incomplete and failed runs.<br>5. The data engineer filters the list or selects a pipeline run.<br>6. The system displays the run details, available error information, provenance and affected data range.<br>7. The data engineer records the issue for follow-up or uses an authorised recovery action where one is implemented.<br>8. The system logs the monitoring or recovery action and refreshes the displayed status. |               |
| <b>Sub-flow:</b> S1. The data engineer may follow a related incorrect-result report by continuing with Use Case 42.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
| <b>Alternative flow:</b> 3a. If monitoring data is stale or incomplete, the system labels its last update time and does not present it as current.<br>3b. If the monitoring service cannot be reached, the system displays an error and offers a retry option.<br>6a. If detailed logs are unavailable, the system displays the available summary and identifies the missing information.<br>7a. If the user lacks authority for a recovery action, the system provides read-only information and records no change.<br>8a. If an authorised recovery action fails, the system preserves the failure state, records the attempt and provides the available error information.                                                                                                            |               |

## 4. Non-functional Requirements

### 4.1 User Capacity (Scalability)

**Concurrent Usage:** The system must be able to support at least 100 concurrent users.

**Peak Traffic Load:** The system must be able to function during peak traffic times and handle 100 requests per minute.

**Scalability:** The system and database architecture must be designed to handle a 20% increase in the user base per year.

### 4.2 Load Time

**Initial UI Load:** The initial user interface of the application, including the home, login and registration screens, must load within 2 seconds over a stable mobile connection.

**Interaction Latency:** User-interface elements must respond to user input, such as clicks or taps, within 1 second.

**Image and Asset Optimisation:** The application should lazy-load images and compress assets to ensure they load in no longer than 2.5 seconds.

### 4.3 Database Storage Space

**Limit:** The system should provide a notification to the user if they exceed 80% of their allocated personal storage quota.

### 4.4 Result Generation Time

**Short-form Text-based Content:** The system should generate results in under 5 minutes.

**Links:** The system should generate results in under 5 minutes.

**Image-related Functions:** The system should generate results in under 10 minutes.

**Timeout:** The system should have a timeout mechanism that cancels processes exceeding 20 minutes to prevent resource exhaustion.

### 4.5 System Security

**Data in the Database and Application:** All sensitive user data stored in the database must be encrypted. All sensitive user data available to system administrators must also be encrypted.

**Authentication:** The system must implement usernames and passwords for all user accounts.

**Session Management:** User sessions should automatically expire after 30 minutes of inactivity.